

MIC Semester III Assignment – Chemistry

Name: _____

Roll No.: _____

Course/Semester: MIC Semester III

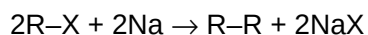
Subject: Chemistry

1. Explain the Mechanism of Wurtz Reaction

Introduction

The Wurtz reaction is an important method for the preparation of higher alkanes. In this reaction, two molecules of an alkyl halide react with metallic sodium in the presence of dry ether to form an alkane containing a larger carbon chain.

General Reaction



where R is an alkyl group and X is Cl, Br or I.

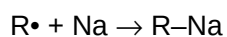
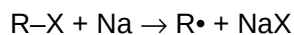
For example:



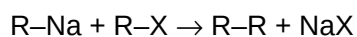
Mechanism

The exact mechanism can involve both radical and organosodium pathways and is commonly represented through formation of an organosodium intermediate.

Step 1: Formation of an alkyl radical/organosodium species



Step 2: Coupling



Thus, two alkyl groups combine to produce the alkane.

Example: Preparation of Butane



The product is n-butane.

Conditions

1. Metallic sodium is used.
2. The reaction is preferably carried out with primary alkyl halides.
3. Dry ether is used as solvent.
4. Moisture must be avoided because sodium reacts vigorously with water.

Limitations

1. It is mainly useful for preparing symmetrical alkanes.

2. Two different alkyl halides generally give a mixture of three alkanes.
3. Secondary and tertiary alkyl halides may undergo competing elimination reactions.
4. The reaction is generally most successful with primary alkyl halides.
5. Highly hindered substrates may give poor yields.

Conclusion

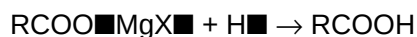
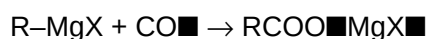
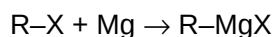
The Wurtz reaction increases the carbon-chain length of an alkane by coupling two alkyl groups using sodium metal. It is particularly useful for the preparation of symmetrical alkanes.

2. Laboratory Synthesis of Carboxylic Acids by Kolbe's Method from a Grignard Reagent

Introduction

Carboxylic acids can be prepared in the laboratory by treating a Grignard reagent with carbon dioxide followed by acidic hydrolysis. This is commonly called carboxylation of a Grignard reagent.

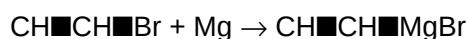
General Reaction



The reaction introduces one carbon atom into the carbon skeleton.

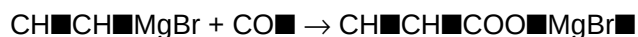
Laboratory Preparation of Propanoic Acid

Step 1: Preparation of the Grignard reagent



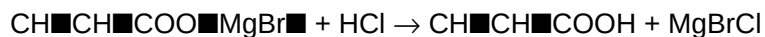
Ethylmagnesium bromide is formed in dry ether.

Step 2: Reaction with carbon dioxide



A magnesium carboxylate is formed.

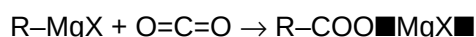
Step 3: Acidic hydrolysis



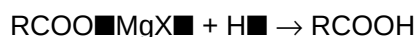
The product is propanoic acid.

Mechanism

The carbon attached to magnesium in a Grignard reagent has nucleophilic character. It attacks the electrophilic carbon atom of carbon dioxide, forming a new C–C bond.



On treatment with dilute acid, the carboxylate ion is protonated:



Important Feature

The carbon chain is increased by exactly one carbon atom because the carbon of CO_2 becomes the carboxyl carbon.

For example:

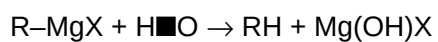


Thus methyl bromide gives ethanoic acid.

Precautions

1. The reaction must be carried out under anhydrous conditions.
2. The ether must be completely dry.

3. Moisture must be excluded because Grignard reagents react with water:



4. Dry carbon dioxide should preferably be used.

5. Acid hydrolysis is performed after reaction with CO_2 is complete.

Advantages

1. It is a convenient laboratory method for preparing carboxylic acids.

2. The carbon chain increases by one carbon atom.

3. Many alkyl groups can be converted into corresponding carboxylic acids if the required Grignard reagent can be prepared.

4. It is useful for preparing acids that may be difficult to obtain by other methods.

Conclusion

Carboxylation of a Grignard reagent with carbon dioxide followed by acidic hydrolysis is an important laboratory method for preparing carboxylic acids.

3. Explain Markovnikov's and Anti-Markovnikov's Addition

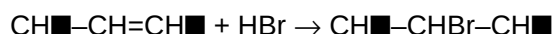
A. Markovnikov's Rule

Statement

Markovnikov's rule states that when an unsymmetrical reagent such as HX adds to an unsymmetrical alkene, the hydrogen atom attaches to the carbon atom that already has more hydrogen atoms, while X attaches to the carbon atom having fewer hydrogen atoms.

In simple words: ****The hydrogen goes to the carbon having more hydrogen.****

Example: Addition of HBr to Propene



The major product is 2-bromopropane.

Mechanism

The reaction generally proceeds through a carbocation intermediate.

Step 1: Protonation



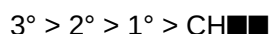
The more stable secondary carbocation is formed.

Step 2: Attack by bromide ion



Reason for Markovnikov Orientation

The orientation is mainly determined by the stability of the carbocation intermediate:



The pathway producing the more stable carbocation is preferred.

B. Anti-Markovnikov Addition

Anti-Markovnikov addition gives the opposite orientation. In the peroxide-induced addition of HBr, the reaction proceeds by a free-radical chain mechanism.

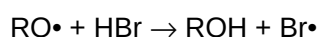
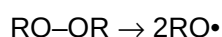
Example



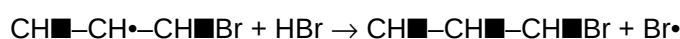
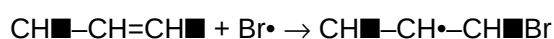
The major product is 1-bromopropane.

Mechanism

Step 1: Initiation



Step 2: Propagation



The bromine radical produced continues the chain.

Comparison

Markovnikov Addition	Anti-Markovnikov Addition
Usually ionic addition	Peroxide-induced HBr addition is radical
Carbocation intermediate generally involved	Carbon radical intermediate involved
Orientation gives Markovnikov product	Orientation is opposite
Carbocation stability controls orientation	Radical stability controls orientation
Example: propene + HBr $\rightarrow$ 2-bromopropane	Propene + HBr/peroxide $\rightarrow$ 1-bromopropane

Important Note

The peroxide effect is characteristic of HBr. It is not generally observed with HCl or HI because the radical propagation steps are energetically unfavorable.

Conclusion

Markovnikov addition and anti-Markovnikov addition differ in the orientation of addition. Normal ionic addition generally follows Markovnikov orientation, whereas peroxide-induced addition of HBr proceeds by a radical mechanism and gives the anti-Markovnikov product.

4. State the Mechanism and Limitations of Free Radical Substitution Reaction

Introduction

A free radical substitution reaction is a reaction in which an atom or group in an organic molecule is replaced by another atom or group through free radicals. The most common example is halogenation of alkanes.

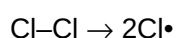
Example:



The reaction proceeds by a chain mechanism consisting of initiation, propagation and termination.

1. Initiation

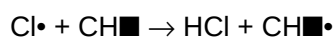
In the presence of ultraviolet light or heat, chlorine undergoes homolytic fission:



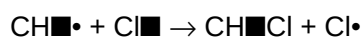
The chlorine atoms are highly reactive free radicals.

2. Propagation

Propagation step 1:



Propagation step 2:



The chlorine radical regenerated in the second step can attack another methane molecule, so the reaction continues as a chain reaction.

3. Termination

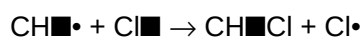
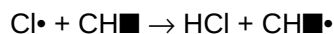
The chain stops when radicals combine.



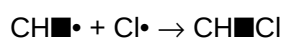
Overall Mechanism

****Initiation:**** $\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$

Propagation:



Termination:



Further Substitution

Chlorination of methane may continue:



Therefore, obtaining only one product can be difficult when excess chlorine and prolonged irradiation are used.

Limitations

1. Mixture of products

When an alkane contains different types of hydrogen atoms, substitution can occur at different positions. For example, propane can give both 1-chloropropane and 2-chloropropane.

2. Difficulty in obtaining a single pure product

Because substitution may occur at different positions, selective preparation of one positional isomer can be difficult.

3. Multiple substitution

After one hydrogen is replaced, further substitution may occur, giving mono-, di-, tri- and tetra-substituted products.

4. Poor selectivity with chlorine

Chlorination is highly reactive and therefore comparatively less selective. Bromination is generally more selective because hydrogen abstraction is more strongly influenced by radical stability.

5. Side reactions

Radicals can combine to form unwanted products. For example:



6. Requirement of suitable conditions

UV light, heat or a suitable radical initiator is generally required.

7. Difficult separation

When several isomeric products are formed, separation may require fractional distillation or chromatography.

8. Not ideal for selective synthesis

Free-radical halogenation is often not preferred when one specific structural isomer is required.

Conclusion

Free-radical substitution is an important reaction of alkanes, especially halogenation. It proceeds through initiation, propagation and termination. Its major limitations are poor selectivity, formation of mixtures of positional isomers, multiple substitution, side reactions and difficulty in obtaining a single pure product.